

Tabu Search C Kodu

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1 Tabu Search Graph Coloring

Tabu search algorithm works as generating solution samples by randomly generating neighbor samples. But, with some limitations, this randomness is organized and leads the algorithm to the exact solution. For the graph $G = (V, E)$ k coloring means that $\forall v \in V$ can be colored with k color so that there is no edge in $e \in E$ adjacent to two same colored vertices.

For graph G a feasible solution will be a partition $s = (V_1, V_2, \dots, V_k)$ of the node set V into a fixed number k of subsets. If $E(V_i)$ is the collection of edges of G with both endpoints in V_i , we may define the objective function f as the number of edges for which both endpoints are in the same V_i (i.e. have the same color):

$$f(s) = \sum (|E(V_i)| : i = 1, \dots, k) \quad (1)$$

In this algorithm, the approach that gives the name of this algorithm is to generate tabu moves at the end of each move (new solution sample). The move means changing the color of one vertex to another color. After every move, the opposite move (changing the old color to that vertex) is added into the cyclic tabu list, and the oldest move in the list is deleted.

For every move, by randomly selecting the vertex x that is adjacent to an edge in $E(V_1) \cup E(V_2) \cup \dots \cup E(V_k)$. For that vertex, we define a move which is the target color so that the color is not the same as the vertex color and $v_i \rightarrow v_j$ move is not in the tabu list. With every move $f(s')$ is calculated and if it is lower than current $f(s)$ then the algorithm directly applies that move and adds it to the tabu list. If $f(s')$ is higher than $f(s)$ then after generating the sample rep by randomly selecting the vertices and changing their colors, the move with minimum $f(s')$ is selected and added to the tabu list.

Also, for generating moves, the aspiration level is used. A function $A(z)$ which is the aspiration level of the objective function value next to be reached when the current value is $z = f(s)$. It is used like this: if a move to a neighbor s' is tabu but gives $f(s') \leq A(z)$, then we drop the tabu status of this move and consider it as a normal member of the sample which is generated. Initially, we set $A(z) = z - 1$ for all values of z . Then whenever we generate an s' with $f(s') \leq A(f(s))$, we set $A(f(s)) = f(s') - 1$.

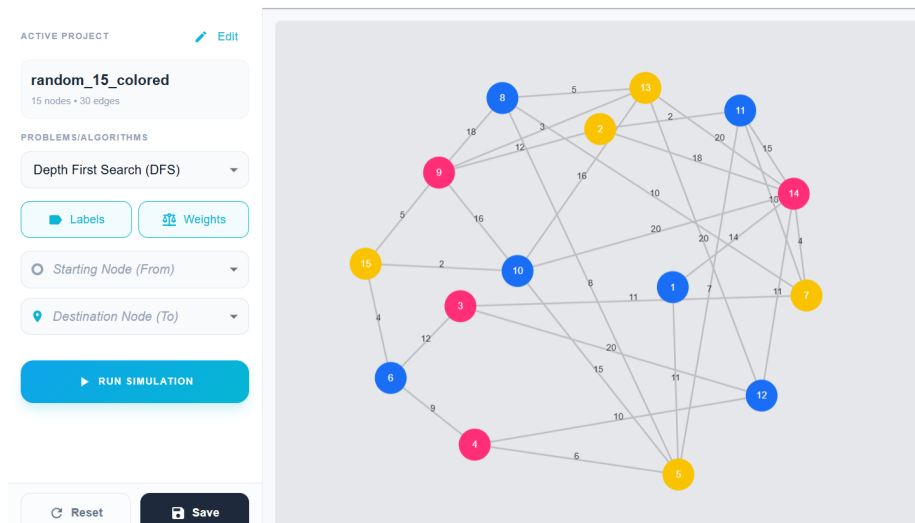


Figure 1: Random 15 graph coloring

2 Test Results

```

:
int main() {
    // your code here
}

```

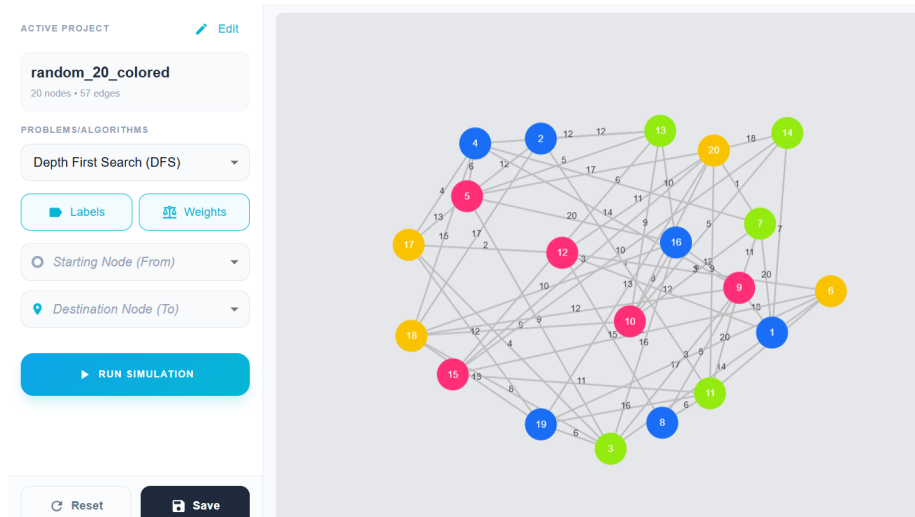


Figure 2: Random 20 graph coloring

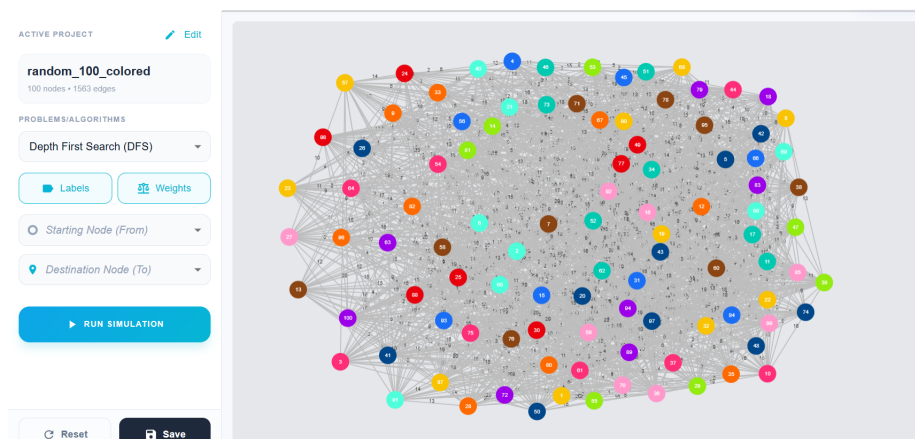


Figure 3: Random 100 graph coloring